

OpenCV: Develop an online writing Whiteboard with minimal features for online classes

AIM

To develop a minimal **Online Writing Whiteboard** using OpenCV in Python that allows users to draw, erase, clear the screen, change colors, and save the whiteboard content.

Introduction

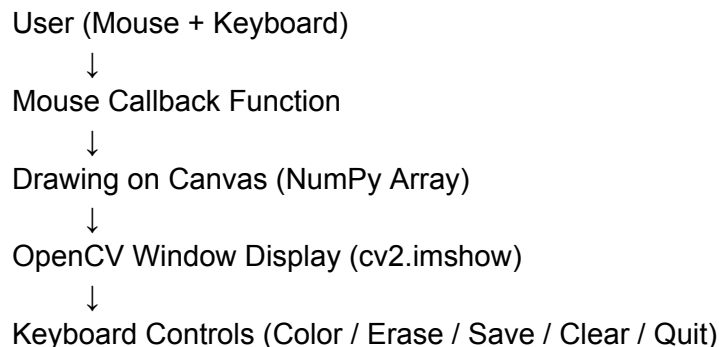
OpenCV (Open Source Computer Vision Library) is an open-source library used for real-time computer vision and image processing. It provides tools for image drawing, video capture, object detection, and GUI handling.

In this experiment, we create a simple online whiteboard using OpenCV. The whiteboard provides a blank canvas where users can draw using the mouse. It supports basic features such as:

- Drawing with mouse
- Changing colors
- Erasing
- Clearing the board
- Adjusting thickness
- Saving the whiteboard as an image

This application is useful for online teaching and demonstrations.

Block Diagram



Software Requirement

- Python 3.8 or higher
- Packages:
 - opencv-python
 - numpy

Install using:

```
pip install opencv-python numpy
```

For Jupyter Notebook, enable GUI backend:

```
%gui tk
```

Procedure

Step 1: Import required libraries (cv2, numpy, datetime).

Step 2: Create a blank white canvas using NumPy.

Step 3: Initialize variables for drawing status, color, thickness, and eraser mode.

Step 4: Define a mouse callback function to:

- Start drawing when left button is pressed
- Draw lines when mouse moves
- Stop drawing when button is released

Step 5: Create a window and attach the mouse callback function.

Step 6: Use a loop to:

- Display the whiteboard
- Capture keyboard inputs
- Change color, thickness, enable eraser
- Clear board
- Save image
- Exit when 'q' is pressed

Step 7: Destroy all windows after exiting.

Full Code :-

Step 1: Import libraries

```
import cv2
import numpy as np
from datetime import datetime
```

Step 2: Create whiteboard canvas

```
CANVAS_HEIGHT = 600
CANVAS_WIDTH = 1000
BACKGROUND_COLOR = (255, 255, 255)
canvas = np.full((CANVAS_HEIGHT, CANVAS_WIDTH, 3),
BACKGROUND_COLOR, dtype=np.uint8)
```

Step 3: Initialize variables

```
drawing = False
ix, iy = -1, -1
color = (0, 0, 0)
thickness = 4
eraser = False
```

Step 4: Mouse callback function

```
def mouse_events(event, x, y, flags, param):
    global drawing, ix, iy, color, thickness, eraser, canvas
    if event == cv2.EVENT_LBUTTONDOWN:
        drawing = True
        ix, iy = x, y
        draw_color = BACKGROUND_COLOR if eraser else color
        cv2.circle(canvas, (x, y), thickness // 2, draw_color, -1)
    elif event == cv2.EVENT_MOUSEMOVE and drawing:
        draw_color = BACKGROUND_COLOR if eraser else color
        cv2.line(canvas, (ix, iy), (x, y), draw_color, thickness)
        ix, iy = x, y
    elif event == cv2.EVENT_LBUTTONUP:
        drawing = False
```

Step 5: Create window and set mouse callback

```
window_name = "OpenCV Whiteboard"
cv2.namedWindow(window_name)
cv2.setMouseCallback(window_name, mouse_events)
print("Controls:")
print("1=Black 2=Red 3=Green 4=Blue")
print("e=Eraser [=Decrease Thickness ]=Increase Thickness")
print("c=Clear s=Save q=Quit")
```

Step 6: Main loop

```
while True:
    cv2.imshow(window_name, canvas)
```

```

key = cv2.waitKey(10) & 0xFF
if key == ord('1'):
    color = (0, 0, 0); eraser = False
elif key == ord('2'):
    color = (0, 0, 255); eraser = False
elif key == ord('3'):
    color = (0, 255, 0); eraser = False
elif key == ord('4'):
    color = (255, 0, 0); eraser = False
elif key == ord('e'):
    eraser = not eraser
elif key == ord('/'):
    thickness = max(1, thickness - 1)
elif key == ord(']'):
    thickness += 1
elif key == ord('c'):
    canvas[:] = BACKGROUND_COLOR
elif key == ord('s'):
    filename = f"whiteboard_{datetime.now().strftime('%Y%m%d_%H%M%S')}.png"
    cv2.imwrite(filename, canvas)
    print("Saved:", filename)
elif key == ord('q') or key == 27:
    break
# Step 7: Close window
cv2.destroyAllWindows()

```

Expected Output

- A window titled “OpenCV Whiteboard” opens.
- User can draw on the white canvas using the mouse.
- Press:
 - 1 → Black color
 - 2 → Red color
 - 3 → Green color
 - 4 → Blue color
 - e → Toggle eraser
 - [/] → Change thickness
 - c → Clear board
 - s → Save image
 - q → Quit

The program successfully creates a simple online whiteboard suitable for online classes and demonstrations.